Tensor

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We begin by defining what we mean by tensor.
A famous definition of tensor is:

> A tensor is something whose components transform like a tensor.

which is cyclic, and not very helpful.

We define it as a mathematical object that faithfully represents the physical quantity of interest, *independent of its coordinate representation*, in the sense that the numbers that represents the tensor may change in a specific way, but the tensor does not.

We will then show this is consistent with the usual transformation rule that defines tensor by levering the power of Dirac's bra-ket. The starting point of all the representation above is that in order to represent the object, the basis vectors must form a *complete set*. We are accustomed to standard basis vector $\{\hat{\mathbf{e}}_i\} \equiv \{\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}\}$, however, a rotated reference is just as valid. We now adopt the new notation the original basis as $\{|i\rangle\}$, which is Latin letters without prime and the new transformed basis as $\{|\alpha'\rangle\}$, which is Greek letters with prime.

Then we put a hat on thermal conductivity $\hat{\kappa}$, much like we put a hat on quantum mechanical operators. However, our thermal conductivity operates in ordinary 3D space.

Then we have the completeness relation,

$$\hat{1} = \sum_i |i\rangle\langle i| = \sum_{\alpha'} |\alpha'\rangle\langle\alpha'| \tag{1}$$

where the sum only goes up to 3.

Then we have,

$$\hat{\kappa} = \sum_{ij} |i\rangle\langle i|\hat{\kappa}|j\rangle\langle j| = \sum_{\alpha'\beta'} |\alpha'\rangle\langle\alpha'|\hat{\kappa}|\beta'\rangle\langle\beta'| \tag{2}$$

which illustrates $\kappa_{ij} \equiv \langle i|\hat{\kappa}|j\rangle$ and $\kappa_{\alpha'\beta'} \equiv \langle\alpha'|\hat{\kappa}|\beta'\rangle$ are really just the different coordinate representation of the same object $\hat{\kappa}$.

Now consider,

$$\langle\alpha'|\hat{\kappa}|\beta'\rangle = \sum_{ij} \langle\alpha'|i\rangle\langle i|\hat{\kappa}|j\rangle\langle j|\beta'\rangle \tag{3}$$

Then defining the transformation matrix $R_{i\alpha'} \equiv \langle i|\alpha'\rangle$, we have,

$$\kappa_{\alpha'\beta'} = \sum_{ij} R_{\alpha' i}\kappa_{ij}R_{j\beta'} = \sum_{ij} R_{\alpha' i}R_{\beta' j}\kappa_{ij} \tag{4}$$

which is the *tensor transformation rule* as we have have claimed.

We have used thermal conductivity tensor as an example, which is a rank-2 tensor, where rank means the number of indicies that the tensor has been decorated with, the above result can be generalised to a more general rank-n tensor.